

CAT-2 AND ASSIGNMENT FOR REAL TIME PROGRAMMING

CAT-2

1. Analyze the role of distributed computing in enhancing the efficiency and reliability of control systems.
2. How do different network communication protocols impact the performance of control systems?
3. Compare the advantages and challenges of using interprocess communication models in networked control systems.
4. Evaluate how POSIX threads handle concurrency issues in networked control applications.
5. Analyze a case study where a networked control system failed due to improper protocol selection. What could have been done differently?
6. Assess the benefits and challenges of implementing networked control systems in smart grid applications.
7. How would you measure the performance of a concurrent programming language in real-time robotics?
8. Evaluate a case study of a successful implementation of POSIX threads in a distributed control system. What factors contributed to its success?
9. Assess the impact of semaphore usage on the responsiveness and efficiency of distributed control applications.
10. Create a concurrent programming solution using POSIX threads for managing multiple autonomous vehicles in a smart city.

ASSIGNMENT -2

11. Create a networked control system for an IoT-based smart home, emphasizing interprocess communication and real-time response.
12. Design a concurrent programming model for a real-time medical monitoring system, addressing synchronization challenges.
13. Propose a novel application of distributed computing for controls in the aerospace industry. Detail the role of communication protocols and concurrency.
14. How does interfacing hardware components in real-time systems impact overall system performance?
15. Analyze the challenges in designing interfaces for real-time programming systems operating under stringent timing constraints.
16. How do real-time scheduling algorithms influence the interfacing of multiple peripherals in a real-time system?
17. Analyze the compatibility issues that arise when integrating legacy hardware with modern real-time programming systems.
18. Examine the role of real-time communication protocols (e.g., CAN, Modbus) in ensuring efficient interfacing in distributed control systems.
19. Analyze how clock synchronization between interfacing devices impacts the accuracy of real-time systems.

20. Investigate the failure modes in hardware-software interfacing and their implications for real-time systems.

ASSIGNMENT -3

21. Evaluate the efficiency of UART versus SPI for interfacing devices in real-time embedded systems.
22. How would you evaluate the performance of I2C bus-based interfacing in multi-sensor real-time systems?
23. Assess the suitability of different interfacing techniques for a real-time medical monitoring system.
24. Evaluate the use of real-time Ethernet for interfacing industrial equipment in smart factories.
25. Assess the trade-offs between using high-level and low-level programming approaches for interfacing in real-time systems.
26. Design a real-time interfacing solution for managing communication between distributed robotic agents.
27. Design an interface for integrating machine learning models into real-time edge computing devices.
28. Develop a system to handle interfacing challenges in a real-time augmented reality application.